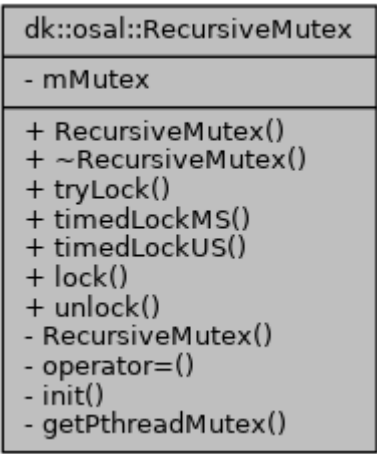


dk::osal::RecursiveMutex Class Reference

OsaiRecursiveMutex shall abstract the recursivemutex functions to lokc the process/thread in recursive manner.
[More...](#)

```
#include <DKOsaiRecursiveMutex.h>
```

Collaboration diagram for dk::osal::RecursiveMutex:



Public Member Functions

- RecursiveMutex ()**
Construct a new **RecursiveMutex**. [More...](#)
- ~RecursiveMutex ()**
- bool tryLock ()**
Try to lock the **RecursiveMutex**, The **RecursiveMutex** is locked if it is not locked by another user. [More...](#)
- bool timedLockMS (uint32_t msec)**
The **RecursiveMutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout milliseconds duration interval elapsed. [More...](#)
- bool timedLockUS (uint32_t usec)**
Lock the **RecursiveMutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout microseconds duration interval elapsed. [More...](#)
- void lock ()**
lock the **RecursiveMutex**. BLocks the current thread until the lock is acquired. [More...](#)
- void unlock ()**
unlocks the **RecursiveMutex**. OsaiMutex component shall provide APIs to unlock the critical section
[More...](#)

Private Member Functions

- RecursiveMutex (const RecursiveMutex &)**
Disable Copy constructor and Assignment operator on **RecursiveMutex**. [More...](#)

RecursiveMutex & **operator=** (const **RecursiveMutex** &)

void **init** ()

Initialize the **RecursiveMutex**. [More...](#)

pthread_mutex_t * **getPthreadMutex** ()

Get pointer to internal pthread_mutex. [More...](#)

Private Attributes

pthread_mutex_t mMutex

Detailed Description

OsaiRecursiveMutex shall abstract the recursivemutex functions to lock the process/thread in recursive manner.

requirement ID 751537

Constructor & Destructor Documentation

◆ RecursiveMutex() [1/2]

dk::osal::RecursiveMutex::RecursiveMutex ()

Construct a new **RecursiveMutex**.

Destruct a **RecursiveMutex**.

requirement ID 751537

Here is the call graph for this function:



◆ ~RecursiveMutex()

dk::osal::RecursiveMutex::~RecursiveMutex ()

◆ RecursiveMutex() [2/2]

dk::osal::RecursiveMutex::RecursiveMutex (const **RecursiveMutex** &)

private

Disable Copy constructor and Assignment operator on **RecursiveMutex**.

Member Function Documentation

◆ getPthreadMutex()

pthread_mutex_t * dk::osal::RecursiveMutex::getPthreadMutex ()

private

Get pointer to internal pthread_mutex.

requirement ID 751534

◆ init()

dk::osal::RecursiveMutex::init ()

private

Initialize the **RecursiveMutex**.

requirement ID 751559

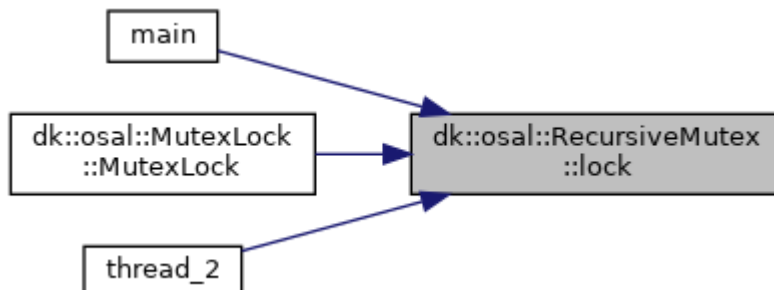
◆ lock()

```
void dk::osal::RecursiveMutex::lock ( )
```

lock the **RecursiveMutex**. Blocks the current thread until the lock is acquired.

requirement ID 751580

Here is the caller graph for this function:



◆ operator=()

```
RecursiveMutex& dk::osal::RecursiveMutex::operator= ( const RecursiveMutex & )
```

private

◆ timedLockMS()

```
dk::osal::RecursiveMutex::timedLockMS ( uint32_t msec )
```

The **RecursiveMutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout milliseconds duration interval elapsed.

Parameters

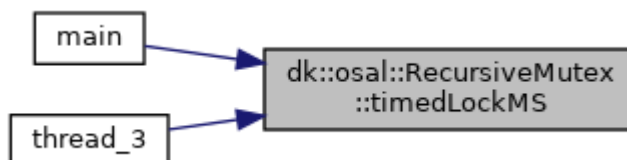
msec The number of milliseconds to wait before the **lock()** is cancelled

Returns

true lock acquired **false** timeout happened

requirement ID 751581

Here is the caller graph for this function:



◆ timedLockUS()

```
dk::osal::RecursiveMutex::timedLockUS ( uint32_t usec )
```

Lock the **RecursiveMutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout microseconds duration interval elapsed.

Parameters

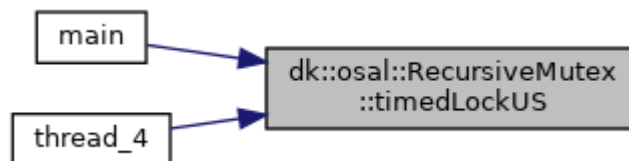
usec The number of microseconds to wait before the **lock()** is cancelled

Returns

true lock acquired **false** timeout happened

requirement ID 751582

Here is the caller graph for this function:



◆ tryLock()

```
dk::osal::RecursiveMutex::tryLock ( )
```

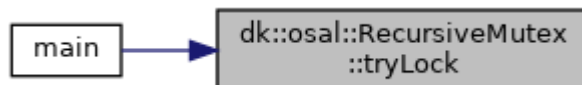
Try to lock the **RecursiveMutex**, The **RecursiveMutex** is locked if it is not locked by another user.

Returns

true lock acquired **false** lock is held by another user.

requirement ID 751580

Here is the caller graph for this function:



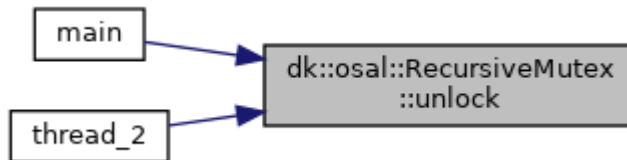
◆ unlock()

```
void dk::osal::RecursiveMutex::unlock ( )
```

unlocks the **RecursiveMutex**. OsalMutex component shall provide APIs to unlock the critical section

requirement ID 751583

Here is the caller graph for this function:



Member Data Documentation

◆ mMutex

pthread_mutex_t dk::osal::RecursiveMutex::mMutex

private

The documentation for this class was generated from the following files:

- `/home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/DKOsalRecursiveMutex.h`
- `/home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/DKOsalRecursiveMutex.cpp`